Formation of Turnbull’s blue

Purpose

To observe the reaction between iron sulfate and potassium ferricyanide and explain the chemical reaction, including the color and electron transfer.

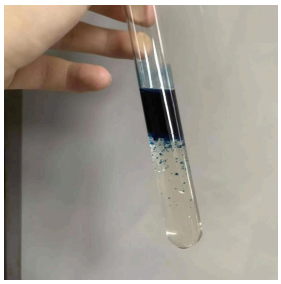
Materials

Iron sulfate solution, potassium ferricyanide solution

Procedure

Combine the iron sulfate and potassium ferricyanide solutions in a test tube.

Observation

Appears in different regions: dark blue at the top, small blue particles at the bottom.	

Chemical explanation

Potassium ferricyanide has a cyanide group that connects iron into a network; the electron transfer changes the iron oxidation states, producing Turnbull's blue, which contains both Fe^{2+} and Fe^{3+}; the dark blue region forms because of intervalence charge transfer. As the light moves electrons around the neighboring iron, which absorbs red light, it turns blue.